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St. Joseph's College of Arts & Science (Autonomous)
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PPH1013 - RESEARCH METHODOLOGY, COMPUTATION METHODS & PROGRAMMING

Time : 3 hrs **Max Marks :75**

SECTION – A (10X2=20)

Answer **ALL** Questions

1. Explain the significance of reference collection.
2. Give any two advantages and cautions to be exercised in seeking internet literature survey.
3. When does Simpson's rule give exact result?
4. Define numerical differentiation.
5. What are called iterates?
6. Which method is suitable to find Eigen values and Eigen vectors for symmetric matrices?
7. What is called forward difference operator?
8. State interpolation.
9. Explain the basic structure of C programming.
10. Define the basic type of C constant.

SECTION – B (5X5=25)

Answer **ALL** Questions

11. a) Describe briefly the art of synopsis writing.

(or)

b) Discuss the usage of multimedia techniques in paper presentation.

12. a) The velocity V of the particle at distances from a point on its path is given by the table:

S	0	10	20	30	40	50	60	feet
V	47	58	64	65	61	52	38	Feet/sec

Estimate the time taken to travel 60 feet by using Simpson's 3/8 rule.

(or)

b) Compute the real root of $x \log_{10} x = 1.2$ correct to three decimal places using Newton's Raphson method.

13. a) How power method can be applied to find numerically the greatest Eigen value of a square matrix.

(or)

b) Explain Eigen values and Eigen vectors.

14. a) Find the least – squares parabola for the four points (-3, 3), (0, 1), (2, 1) and (4, 3).

(or)

b) Construct a forward difference table from the following values of x and y. Show that the third differences are practically constant.

x	**35**	**36**	**37**	**38**	**39**	**40**	**41**
y	**14.298**	**14.144**	**13.986**	**13.825**	**13.661**	**13.495**	**13.328**

15. a) Describe the arithmetic operators on character in C.

(or)

b) Explain one dimensional arrays with examples.

SECTION- C (3X10=30)

Answer any **THREE** Questions

16. Explain in detail the art of writing result and discussion and conclusion part in a thesis.

17. Find the values of y(1.1) using Runge-Kutta method of the third order. Given that $x_0 = 1$, $y_0 = 1$ and $h = 0.1$.

18. Solve the following system by Gaussian elimination method

$x_1 - x_2 + x_3 = 1$

$-3x_1 + 2x_2 - 3x_3 = -6$

$2x_1 - 5x_2 + 3x_3 = 5$

19. Using Gauss's interpolation formula, obtain f(3.5) from the following table.

x	2	3	4	5
F(x)	2.626	3.454	4.784	6.986

20. With suitable examples explain the increment, decrement and conditional operators in C.

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